Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.2 – Match the Graph: Motion Sensor Lab

Unit 2: Forces and Motion

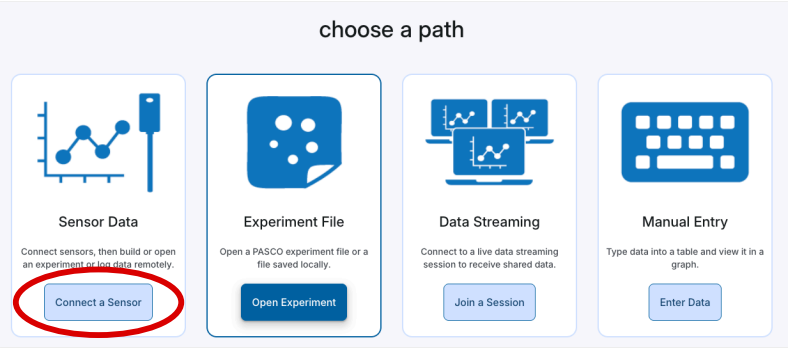
1. Open SPARKvue

Take out your **laptop**, open **Google Chrome**, and go to:

`sparkvue.pasco.com`

2. Connect the sensor

1. Put the sensor on a table, its **round screen at waist height**, facing **3–4 m** of clear floor.
2. Press the **power button**. The light **blinks red**.
3. In SPARKvue, on the **choose a path** screen, click **Connect a Sensor** (ringed), then pick the **number printed on your sensor's label**.

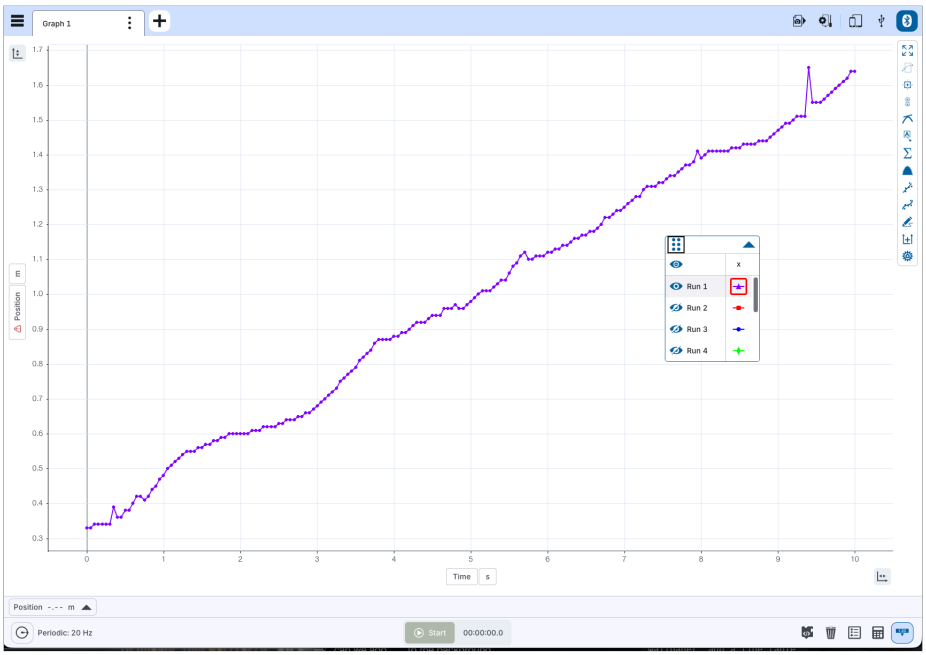


Check: the light now **blinks green**.

3. Show your position

1. In the list of measurements, click **Position (m)**.
2. Click **Graph**.

Check: a graph appears with **Position (m)** up the side and **Time (s)** along the bottom. It **rescales itself** to fit your walk, so read the numbers on *your* axes.

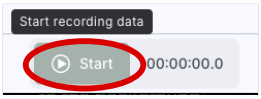


A position graph after a walk.

4. Walk a graph

The sensor measures **your distance from its round screen**. Walk **away** and the line goes **up**; walk **towards** it and the line goes **down**; stand still and it stays **flat**. Stay at least **0.5 m** away.

1. **Plan.** Pick a graph from the next pages. Read where its line starts and decide where to stand and how to move. Use the numbers: a line starting at 1.5 m means you start 1.5 m from the sensor.
2. **Stand** at the starting distance, facing the sensor.
3. **Walk it.** Your partner clicks **Start** (*Start recording data*), bottom of the screen. Move the way you planned, then your partner clicks **Stop** after **8 s**.



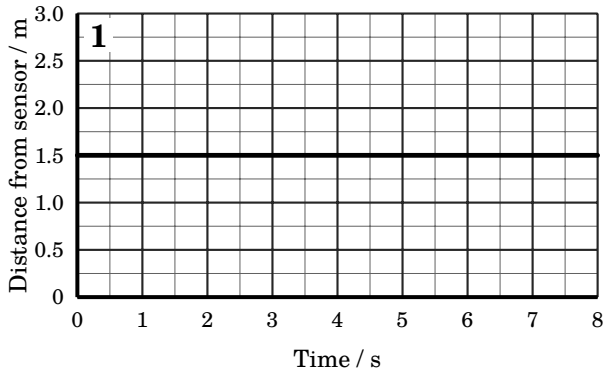
4. **Compare.** Read the numbers on your axes and hold your line against the graph on the sheet: same start distance, same steepness, same turns? Adjust the plan and walk it again until they match.
5. **Write it down.** Under the graph, describe how you moved, using the words in the box. Tick *Matched it?* when your line fits.

Check: everyone walks at least two graphs. **Care:** keep the floor in front of the sensor clear, and walk.

Words to use: stationary • constant speed • away from • towards • faster • slower • speeding up • slowing down

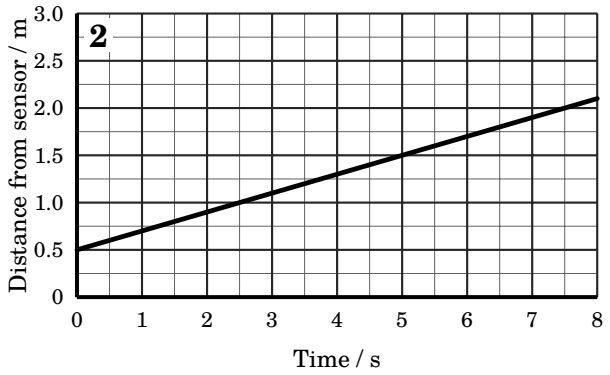
The six graphs

Work through them in order. Number 5 has four parts — describe each one.



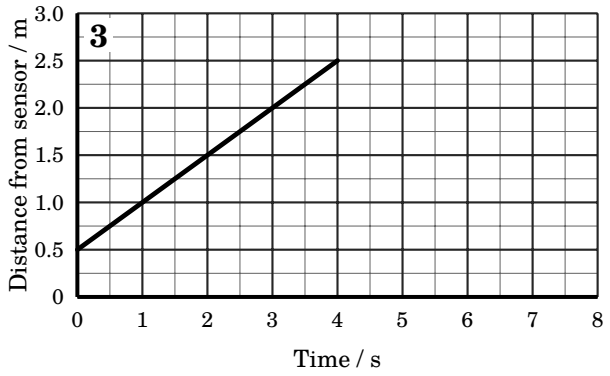
How I moved: **Matched it?** ☐

Stood still (stationary), 1.5 m from the sensor, for all 8 s.



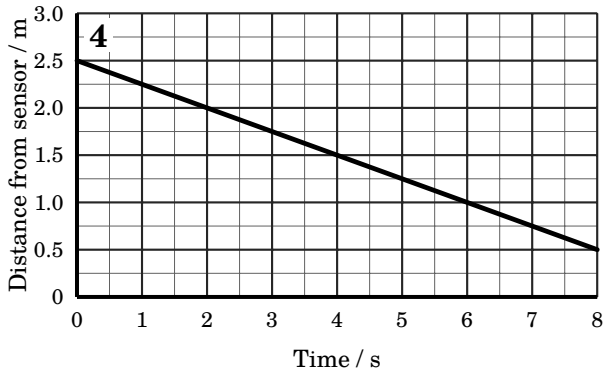
How I moved: **Matched it?** ☐

Walked away from the sensor slowly, at a constant speed, from 0.5 m to 2.1 m.



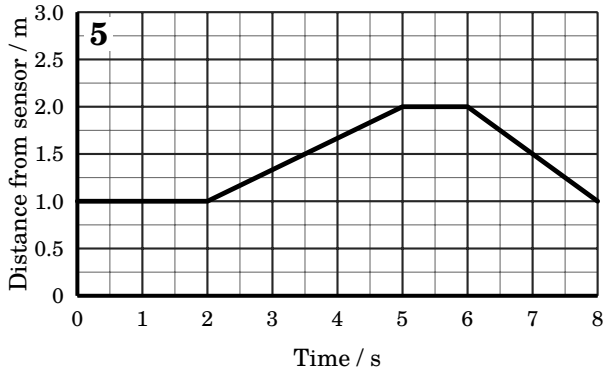
How I moved: **Matched it?** ☐

Walked away at a constant speed, faster than in 2: from 0.5 m to 2.5 m in only 4 s. The line is steeper.



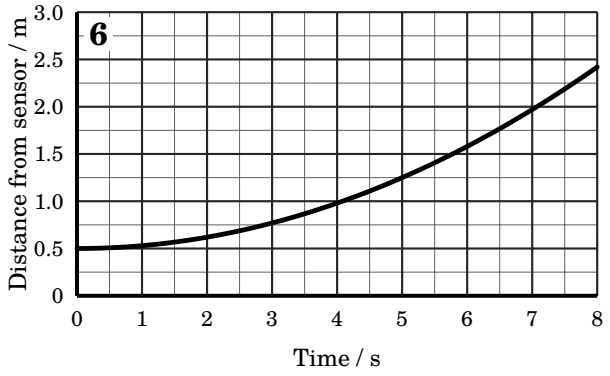
How I moved: **Matched it?** ☐

Walked towards the sensor at a constant speed, from 2.5 m to 0.5 m.



How I moved: **Matched it?** ☐

Stationary at 1.0 m for 2 s; walked away at a constant speed to 2.0 m; stationary for 1 s; walked back towards the sensor to 1.0 m, faster than on the way out.



How I moved: **Matched it?** ☐

Walked away from the sensor, speeding up: slow steps at first, getting faster.

What the graphs show

1. On a distance–time graph, a flat line means the walker is **stationary**.

2. Graphs 2 and 3 are both straight lines going up. Which one was the faster walk, and how can you tell from the graph?

Model answer: Graph 3, because its line is steeper: it covers more distance in each second.

3. Use Graph 3 to work out the walker's speed. They moved from 0.5 m to 2.5 m in 4 s.

speed = distance ÷ time = **2.0** m ÷ **4** s = **0.5** m/s

Check: the distance is how far they *moved*, end minus start.

To go further — the impossible graph

Nobody in the room can walk a graph that goes **straight up**, a vertical line. Explain why, using what a vertical line would mean about distance and time.

Model answer: A vertical line means the distance changes while no time passes at all. That would need an infinite speed, so no one can move from one place to another in zero time.